Nasan-Ochir Sukhbaatar

Applied AI Engineer / Tech Lead / Full-Stack and Mobile

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SUMMARY

Applied AI engineer and team lead with 8+ years shipping software in Mongolia across government, banking and consumer products. I train and ship models for a low-resource language: my fine-tuned Mongolian voice model dubs full drama episodes with distinct, emotional native voices. I also build grounded LLM products end to end, from data pipelines to evaluation and paid production use. Before that I led mobile teams on a national e-library and an 8,500-seat arena platform, built bank apps, and shipped loyalty, ordering and enterprise apps for GS25, APU and Shunkhlai Group.

SELECTED AI WORK

- **Mongolian drama dubbing model (trained myself).** Built a ~92-hour, 95-speaker Mongolian speech corpus and fine-tuned a voice model on it, including emotion training (angry, sad, tender delivery). Built per-character voice casting that keeps one consistent voice per character across a series, and an automated QA gate that checks every line for correct words, voice, gender, pitch, timing and a model-based listening test that matches human review 87-93% of the time. Built the subtitle side too: it reads the source subtitles (official tracks, or burned-in text read from the video frames), rewrites them as natural Mongolian sized to each line's timing, and burns Mongolian subtitles into the final episode. **Watch: nasanochir.github.io/mongolian-ai-dubbing**
- **Grounded civic AI (Open Parliament, 3rd place).** Search and chat over parliament agendas, transcripts and 950 laws (15k articles), built only on official open government data. Missing data is shown as a gap, never filled in by the model. Facts are kept only when backed by a verbatim official quote. A validation rule threw out 345 of 550 wrong document links. The assistant re-answers under a stricter grounding rule when it fails to cite sources.
- **Paid LLM product (astro.mn).** 14 personalized long-form reading types and a chat that remembers the user. Facts are computed in code and locked into the prompt so the model can't invent numbers. Live with paid checkout.
- **Multi-IQ customer service AI agent (Microsoft EMEA Partner Hackathon, 2026, L300 track).** Built a Copilot Studio agent, live in Microsoft Teams, that combines grounded policy (Azure AI Search, Microsoft Foundry), order and SLA data, and expert routing (Azure Functions) to resolve a rep's escalation in one 24-second answer, identical across repeat runs. Every escalation is logged to a 12-field Dataverse governance table with a human approval written back, shown in a Power Apps audit console. Refused 3 of 3 prompt-injection attacks.
- **Mongolian language tools.** Mongolian meeting transcription and AI meeting notes SaaS (ZeroTech). A spell checker with 96% top-1 suggestion accuracy against 72% for the market leader, on a held-out benchmark.
- **AI adoption.** Designed Shunkhlai Group's 4-module employee AI training program. Use multi-agent coding workflows day to day.

EXPERIENCE

ZeroTech LLC

Jul 2023 - Oct 2026

Senior Software Developer (mobile, web, backend)

Campus super-app (main developer) Flutter, Riverpod, Firebase, WebView

- Built the app from scratch, including a host for web mini-apps (food ordering, meeting rooms, facility, events, parking) that passes login tokens and handles navigation, and an indoor wayfinding map with a building directory.

GS25 Mongolia: loyalty app, CRM admin, e-receipt panel Flutter, Riverpod, WebSockets, Next.js, Refine, Mantine, Medusa UI, Docker

- Built the retail loyalty app from its first version (eBarimt e-receipts, vouchers, coupons, in-app shop, wallet passes, WebSockets) and the loyalty/CRM admin dashboards. Top contributor to the e-receipt panel: batch sending, eBarimt matching, SSO and roles, Docker CI/CD.

DOT B2B ordering apps (top contributor) Flutter, Riverpod, Provider, Freezed, Firebase

- Started and built the customer ordering app (catalog, per-company cart, discounts, order templates, branch management, finance statements, asset and fridge tracking, push notifications) and the sales-rep ordering app.

APU Dairy and APU Route React Native / Expo, Turborepo, Next.js, Drizzle ORM, AG Grid

- Started and built the APU Dairy milk-collection app (procurement, driver and lab workflows, offline sync, Bluetooth receipt printing) and built the APU Route fuel accounting system (fuel calculations, vehicles, Excel reports).

Car parking admin panel and POS monitoring React, Node.js, PostgreSQL, Drizzle ORM, TanStack, Docker

- Restructured the parking admin panel per the PRD and built reports with CSV/PDF export, settlement reconciliation, user groups with CSV import, pricing setup and the payment front end. Built a POS fleet monitoring dashboard.

goOrange: Microsoft Teams peer-recognition app (3rd place, Shunkhlai Hackathon) React 19, Supabase (Postgres, Edge Functions), Bot Framework, Microsoft Graph, MCP

- Built a Teams tab and bot for peer recognition: weekly allowances, rewards redemption, idea board, Graph-synced directory, proactive notifications and an MCP/agent interface.

Shunkhlai Group apps (contributor): e-Mongolia and QPay integration, notifications and loyalty features in the Orange consumer app (Flutter); delivery history in the B2B driver app.

SteppeLink LLC (Steppe Link Holding)

Dec 2021 - Jul 2023

Team Lead, Mobile (team of about 5)

Esan: national digital library iOS, Android, Readium

- Project lead with a team of 8 on Mongolia's national e-library (ebooks, audiobooks, theses, newspapers, magazines) with offline reading.
- Integrated the Readium SDK for EPUB rendering, with pagination, highlights and reading-position sync.

Steppe Arena app

- Delivered the app for the 8,500-seat AIC Steppe Arena: event and sports ticketing, in-seat food ordering, ice-skating session booking, and plate-based smart-parking payment.

Odin Survey

- Built a survey and market-research platform (web, iOS, Android) where users earn points and coupons for completing surveys and business clients get aggregated insights.

Infinite Solutions LLC

Oct 2018 - Dec 2021

Senior Software Developer, Mobile

- Built and maintained mobile banking apps for major Mongolian banks: **Arig Bank, XacBank, Golomt Bank, Trade and Development Bank (TDB)**. Work covered secure login, accounts, transfers, card management and payments.
- Developed consumer apps **Shuffle** and **Elify**.
- Worked within a fintech vendor building payments, merchant and loyalty infrastructure under banking security and compliance requirements.

Astvision LLC

Apr 2018 - Oct 2018

Software Developer, Full-Stack

- Full-stack work on government e-service systems: early **e-Mongolia** platform work, the Ulaanbaatar city e-services portal (**e-service.ulaanbaatar.mn**), and **my.tog.mn**, the Ulaanbaatar Electricity Distribution Network customer portal for bills, consumption and service requests.

INDEPENDENT PRODUCTS AND AWARDS

- **3rd place, Open Parliament Hackathon (2026)**, organized by the Secretariat of the State Great Khural, The Asia Foundation and Unread Media. 24-hour build of an MP accountability platform (votes, attendance, speeches, bill compliance, grounded AI chat).
- **kinody.mn**: mobile-first Asian drama streaming (97% mobile audience) with QPay subscriptions. Home of the dubbing model.
- **gadaad.mn**: visa helper for 10 countries: passport-scan autofill, document checklists, paid AI approval scoring.
- **Koharu** (open source, MIT): contributor to a desktop manga image translator (Rust, Tauri 2, local LLMs).

EDUCATION

Sapienza University of Rome, M.Eng. program in Computer Science and AI (completed first year) 2017 - 2018

National University of Mongolia, Bachelor of Information Technology 2013 - 2017

SKILLS

Mobile	Swift, Objective-C, Kotlin, Java (native iOS/Android), Flutter/Dart (Riverpod, Freezed), React Native/Expo
Web	Next.js (App Router, 13-16), React 19, TypeScript, Tailwind, Mantine, TanStack Query, Zustand
Backend	Node.js (Express, TSOA, tRPC), Python (FastAPI), SQL (PostgreSQL), Supabase (RLS, Edge Functions), Drizzle, Redis, Docker
ML / Speech	Python (PyTorch, scikit-learn, pandas), model fine-tuning, speech corpus building, TTS, speech-to-text, evaluation and QA gates, GPU training
LLM Eng.	RAG and embeddings, grounding and fact validation, agents and MCP, streaming, prompt and cost optimization (Claude, OpenAI, Gemini)
Integrations	QPay, eBarimt e-receipts, bank payment gateways, e-Mongolia XYP, Microsoft Graph/Teams, Bluetooth printing
Microsoft AI	Copilot Studio, Microsoft Foundry, Azure OpenAI, AI Search, AI Speech, AI Language, Functions, Dataverse
Cloud	Vercel, Cloudflare Workers, Azure, GCP, Linux GPU servers (RunPod)
Languages	Mongolian (native), English (professional)